## DATA 3464: Fundamentals of Data Processing

# Splitting and combining data

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# Topic overview

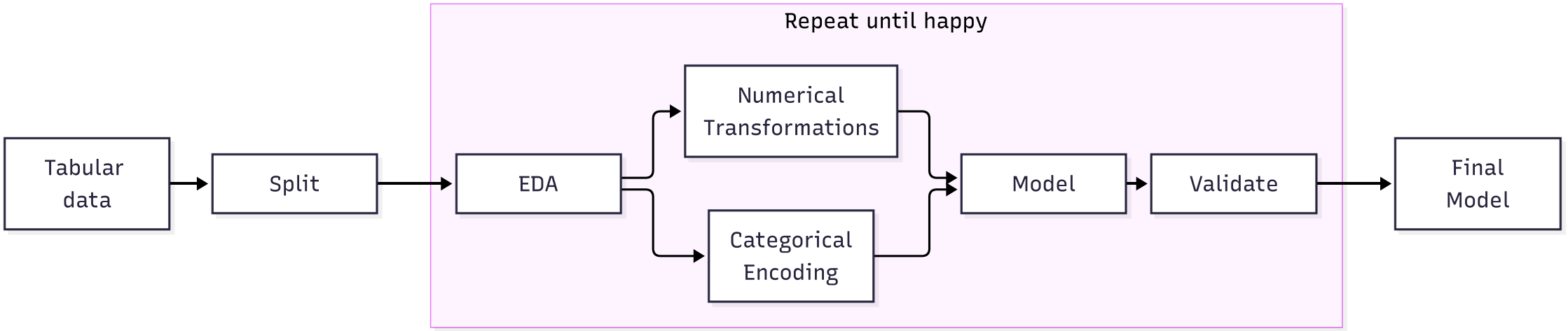
Adding detail to the process thus far

- Splitting and sampling data
- Some more visualization tricks
- Combining multiple datasets

**Resources used:**

- Feature Engineering Sections 3.3-3.4
- Hands on Machine Learning with Scikit-Learn and Tensorflow/PyTorch, Chapter 2. Available at MRU Library

# The data process to date




- We've talked about *why* to split, but not *how*
- We might also have more than one table to deal with

# How to split your data

- Simple scenario: random sample (typically 70-80% for training)
- **Only works if:**
  - Stratification doesn't matter
  - Data is guaranteed not to change
  - Data is not a time-series

> *How do we know if stratification is necessary?*

# Sampling bias

Stratification is used to mitigate **sampling bias**

- Cilantro example: assume 80% of population likes cilantro
- Goal: ensure our sample is representative of the population, $\pm 5\%$

The binomial distribution can be used to model the probability of choosing $k$ people who like cilantro from $n$ total participants:

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}, \text{where} \binom{n}{k} = \frac{n!}{k!(n-k)!}$$

# Sampling bias continued

$P(X = k)$ is the probability mass function, and the corresponding cumulative distribution function is just the sum up to $k$:

$$P(X \leq k) = \sum_{i=0}^{k} \binom{n}{i} p^i (1-p)^{n-i}$$

Suppose we **randomly** sample 100 people. What is the probability of fewer than ~~75~~ 79 or more than ~~85~~ 81 cilantro lovers?

> *Here I've defined an "unbiased sample" as being $\pm 5\%$ of expected **of the smallest category** (that this is a correction from the version presented in class!)*

# Stratification approach

- The need for stratification depends on sample size, distribution of stratification category, and how much bias you're willing to accept

| | **Small Sample Size** | **Large Sample Size** |
|---|---|---|
| Unbalanced Classes | Stratify | Maybe |
| Balanced Classes | Maybe | Not necessary |

- Stratification categories can be the target variable, or a predictor
- Goal is to have the same class distribution in both testing and training

# Repeatable randomness

- At minimum, you should **always set a random seed** so that every time you sample your data it is the same "random sample"
- This isn't enough if your data might get updated! You can:
  - Store the IDs of your split offline, then sample any new data and append to them (ensuring that test/train never mix)
  - Get fancy with a deterministic method like hashing features to create unique IDs, then thresholding based on maximum possible value

> *As usual, we want to avoid **data leakage***

# Back to visualizations

Now that we've got a test set safely stashed, we can **ask questions** about the data and use visualizations and statistics to answer them. Some examples:

- Do any of my features seem to be related to my target?
- Do any of my features seem to be related to each other?
- Why are some values more common than others?
- Do these values make sense in the context of my **domain knowledge**?
- If I group my data together in some way, are there clear trends?

R for Data Science has some good examples of further questions

# Some handy tricks

A few things to tweak that can make visualizations easier to read:

- Histogram bin sizes
  - Aiming for a smooth distribution that works for your data
- Transparency (`alpha`)
  - Useful for both dense scatter plots and overlapping categories
- "Jitter"
  - Mostly for scatter plot of continuous vs categorical data
  - Add a tiny bit of random noise to spread out samples

# Combining data sources

My assumption: you know all about joins from DATA 2721 and DATA 2402

- Concatenating tables: `pd.concat([list, of, dataframes], axis=??)`
- Joining to *nearest* match: `pd.asof`
- Watch out for datatypes! Dates in particular are a huge pain
  - `datetime[ns]` is not the same as `datetime[s]`

# Coming up next

- No lab this week (Truth and Reconciliation)
- Assignment 1 is due!
- Next topic: Missing data and outliers
- Written test 1 next **Thursday**